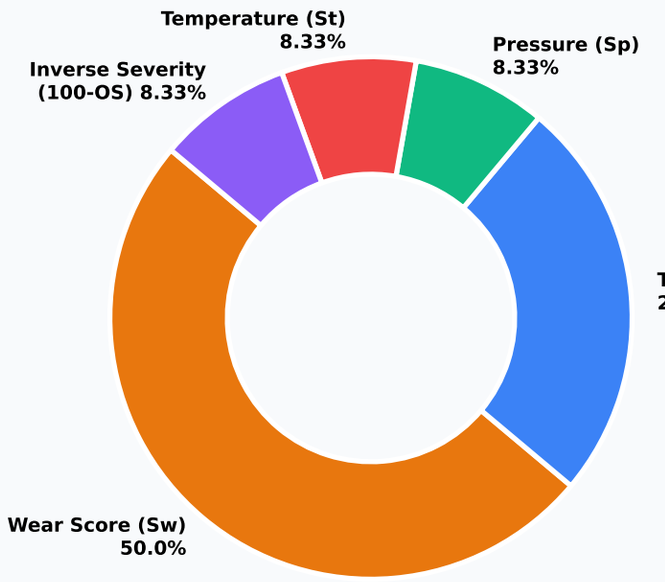
TYRE HEALTH INDEX (THI) – EXECUTIVE SUMMARY & DATASET ARCHITECTURE

Formula-Based Tyre Health Assessment Across 3 Closed Periodic Inspections & High-Frequency Telematics

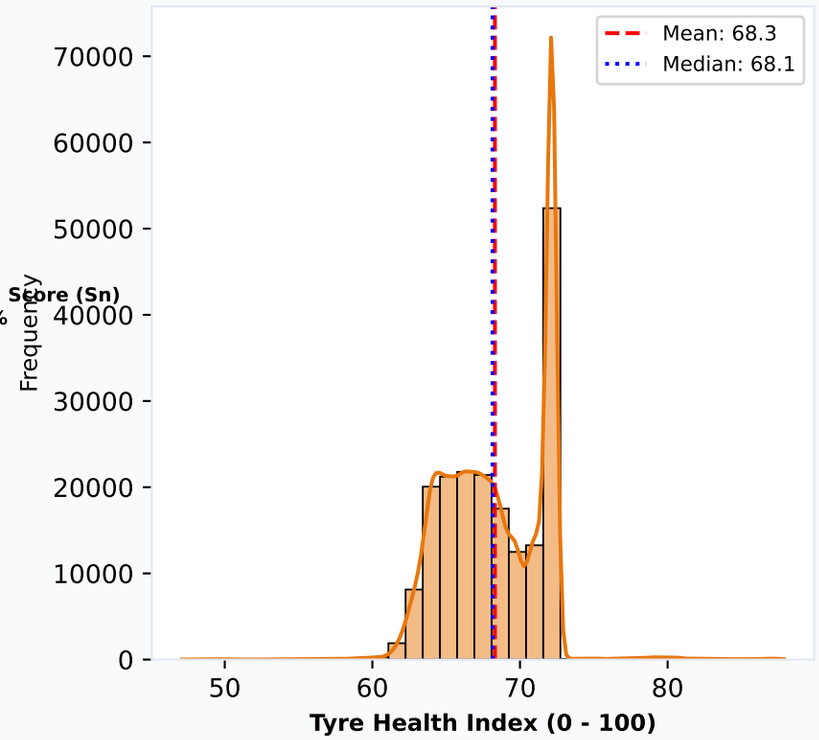
Dataset & Fleet Layout Metrics

Operational Metric	Assessed Value
Total Telematics Records	561,605 records (6s cadence)
Sampling Window	18 Oct 2025 – 25 Nov 2025
Periodic Inspections	2nd, 3rd, and 4th Closed Audits
Sampled ML Observations	192,540 instances
Fitted Wheel Positions	12 Active Wheels + 1 Spare
Axle Distribution	Steer (2), Drive (2), Tandem (8)
Target Vehicle OTD	16.5 mm (JUH5 REG/EXP)
Mean Recorded Speed	30.7 km/h (max 80.0 km/h)
Mean Tyre Pressure	126.2 PSI (std 2.25)
Mean Operating Temp	34.7 °C (22.4 – 45.0 °C)

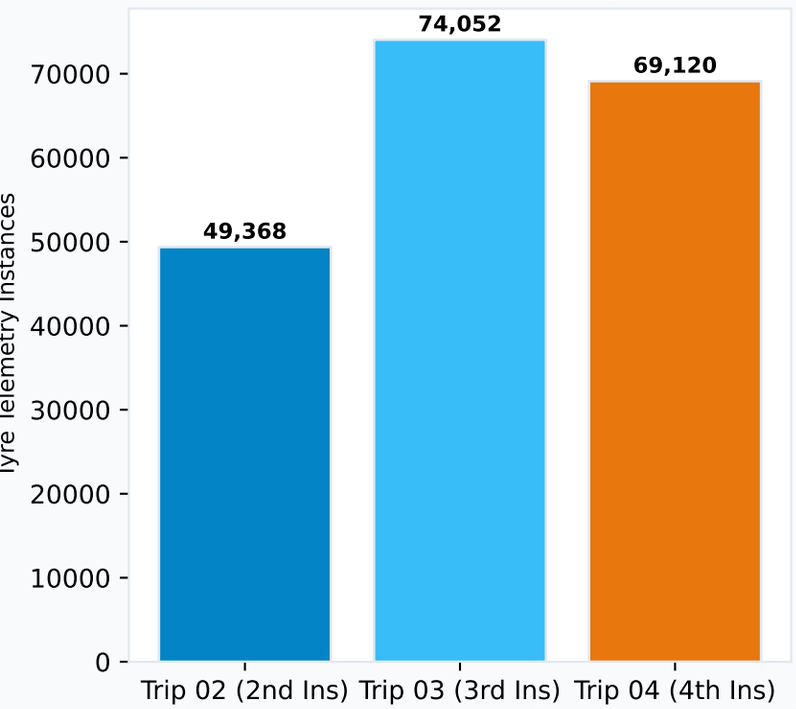
Executive THI Weight Architecture



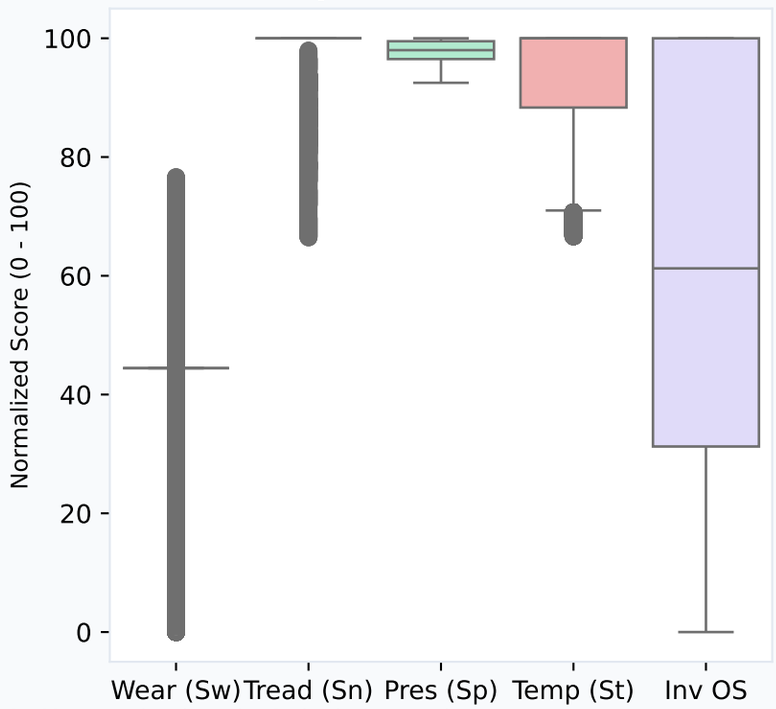
Overall THI Score Distribution



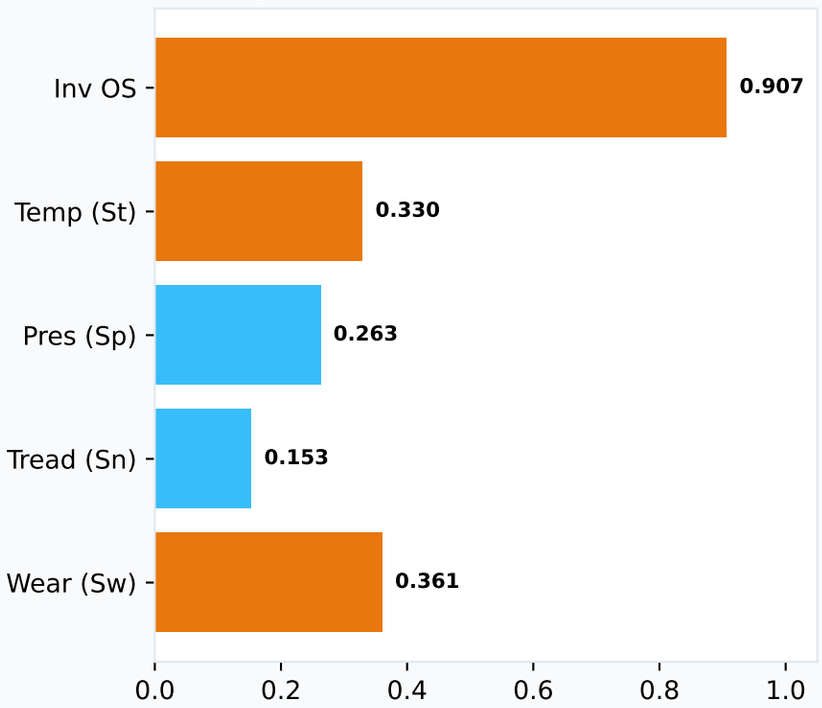
Sampled Observations Per Inspection Trip



THI Component Score Variations



Component Correlation with Final THI

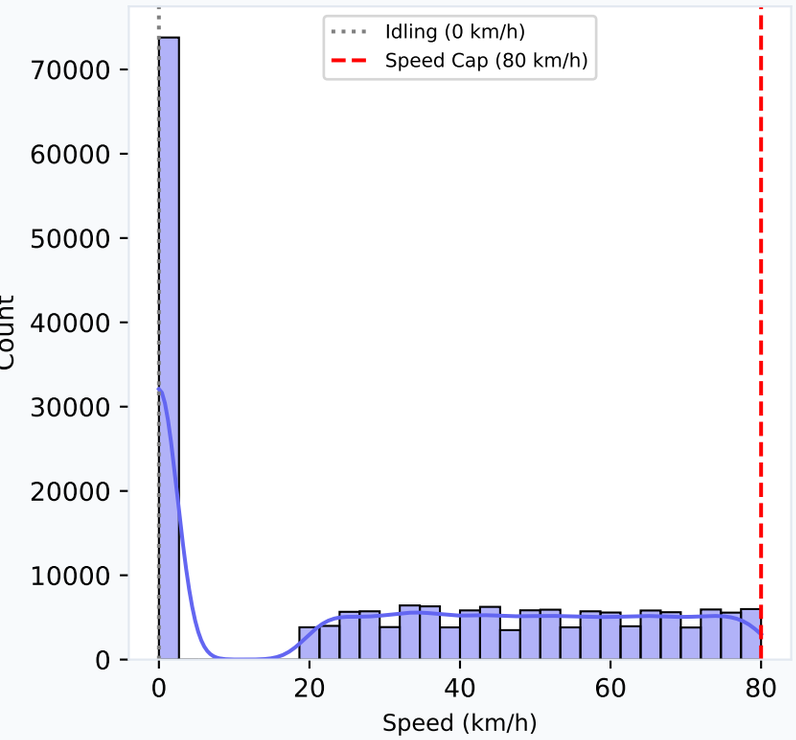


Formula: THI = 0.50*Sw + 0.25*Sn + (0.25/3)*(Sp + St + 100-OS)

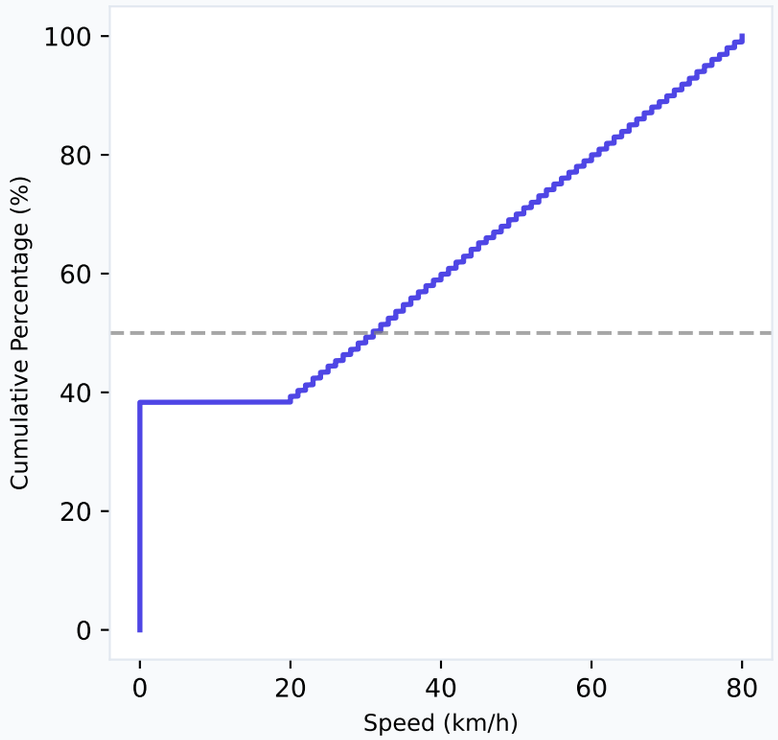
TELEMATICS SENSOR EXPLORATION: SPEED, PRESSURE & TEMPERATURE

Empirical Sensor Distributions Across 561,605 Telematics Records (18 Oct – 25 Nov 2025)

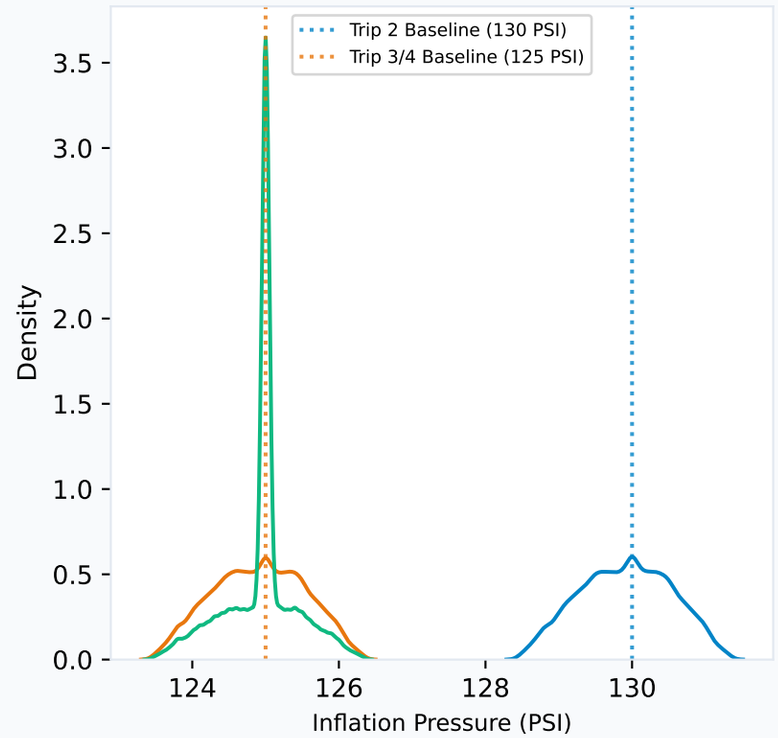
Vehicle Speed Dynamics (km/h)



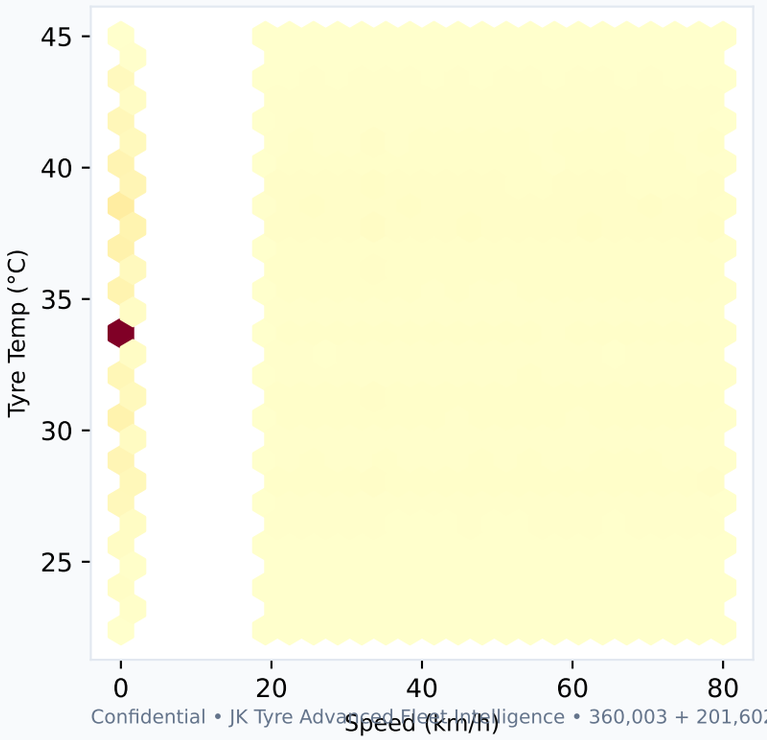
Operating Regime CDF (Speed Exposure)



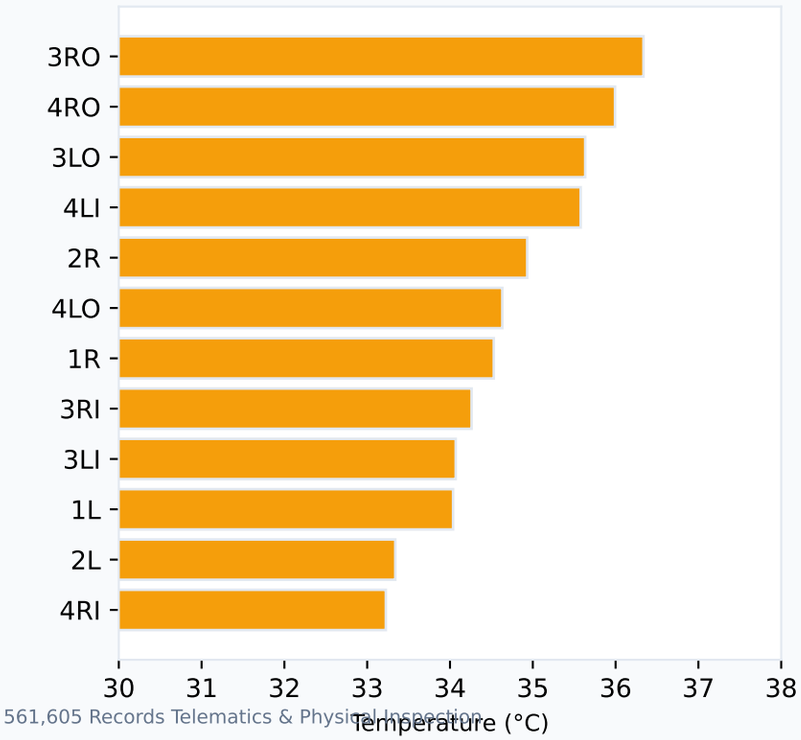
Tyre Pressure Profiles Across Trips (PSI)



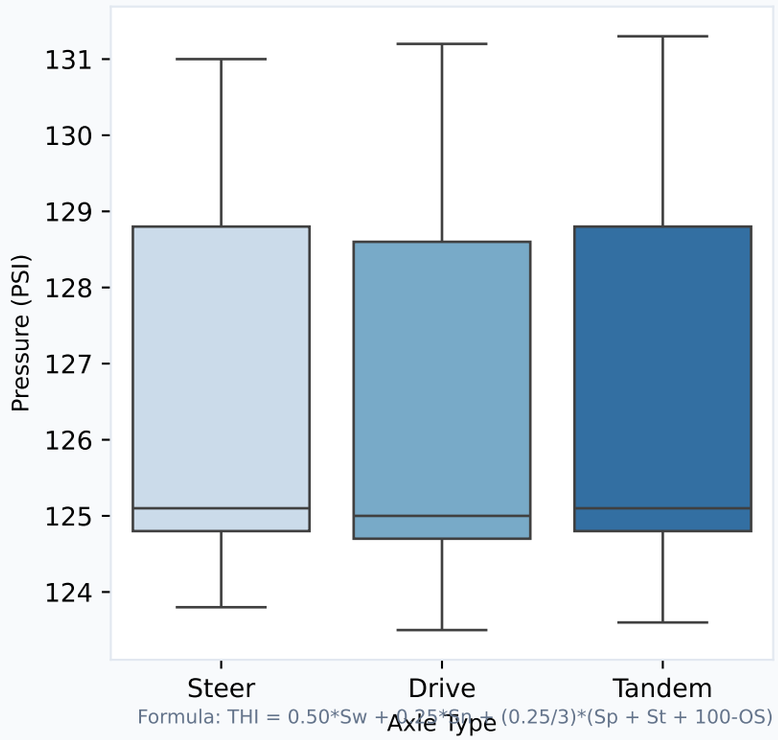
Speed vs Tyre Operating Temp (°C)



Mean Operating Temp by Position (°C)



Pressure Consistency Across Axles

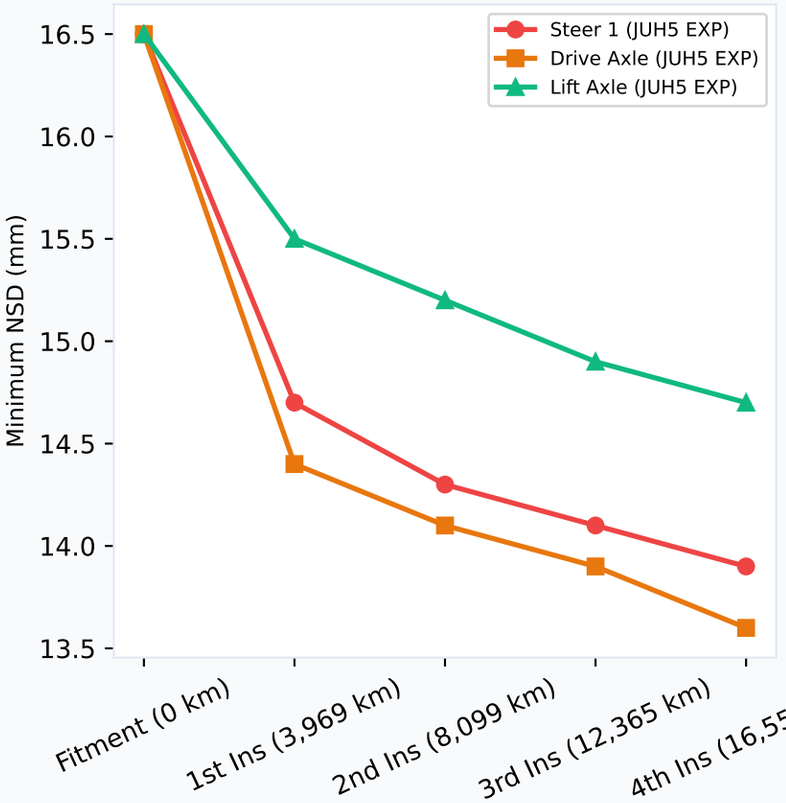


Formula: THI = 0.50*Sw + 0.25*Sp + (0.25/3)*(Sp + St + 100-OS)

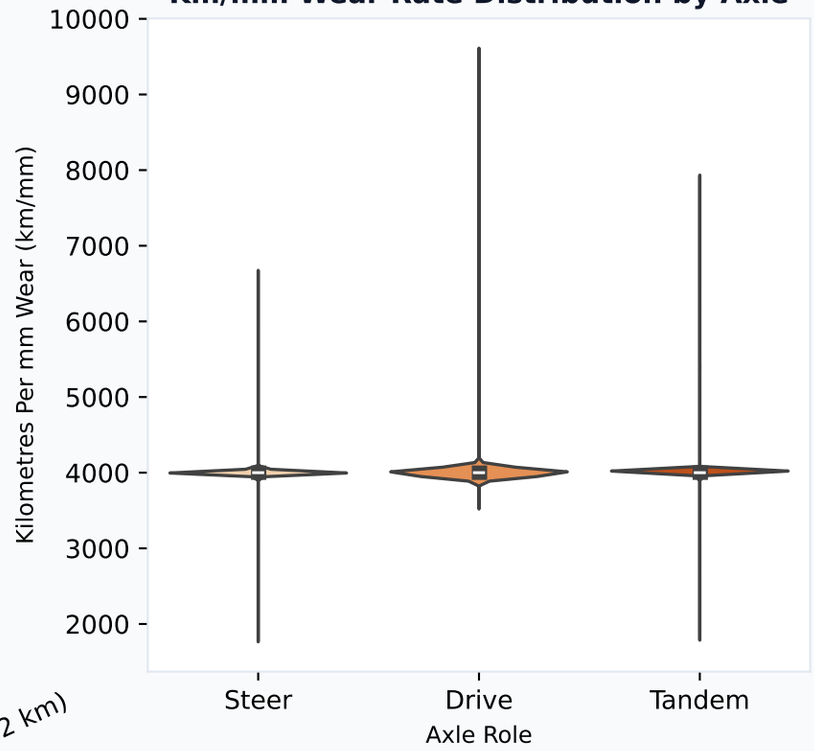
PHYSICAL INSPECTION & TREAD WEAR PROGRESSION ANALYSIS

NSD Degradation, Km/mm Wear Rates & Projected Mileage Across 4 Periodic Inspection Milestones

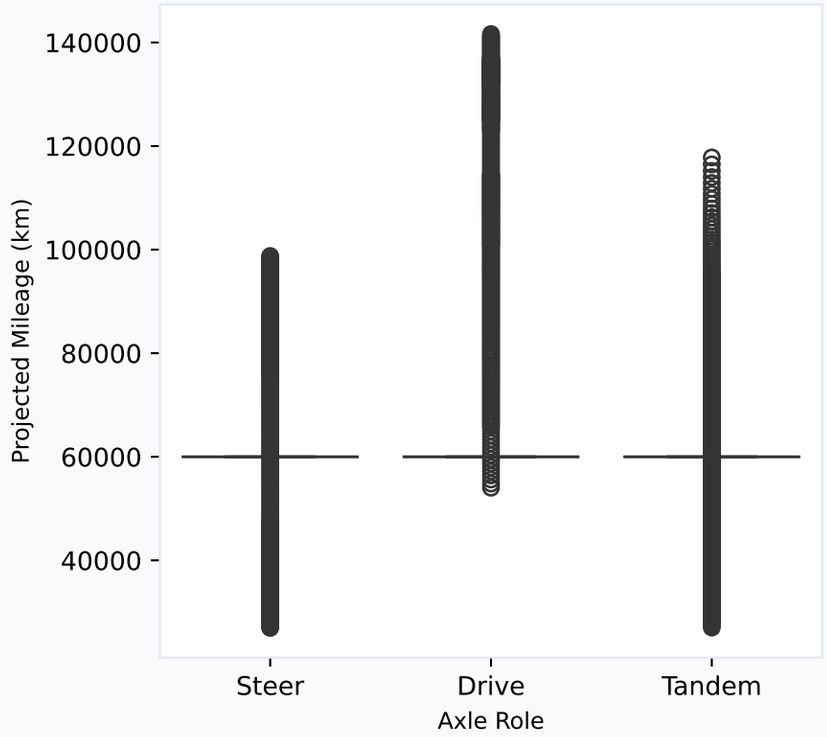
NSD Tread Wear Progression (mm)



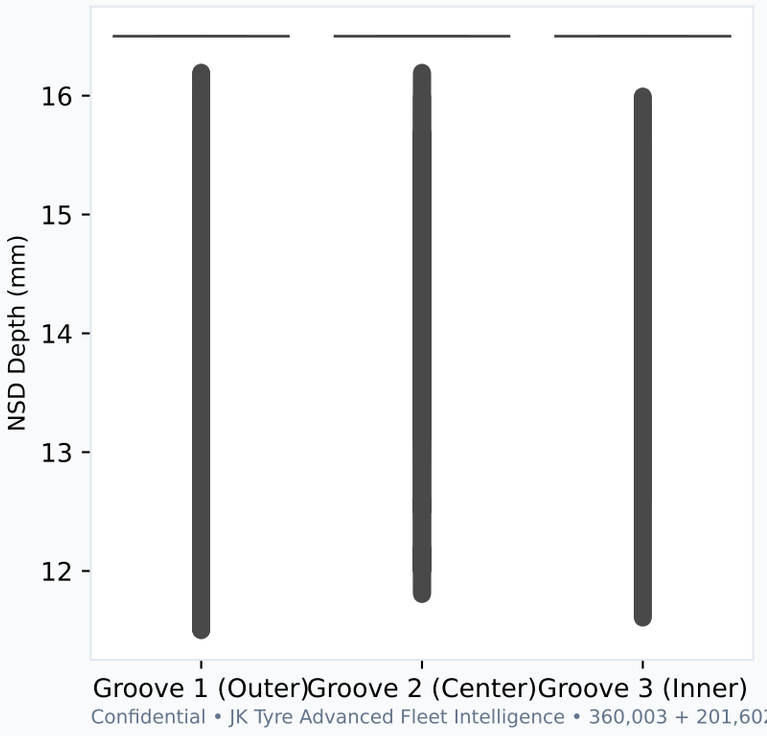
Km/mm Wear Rate Distribution by Axle



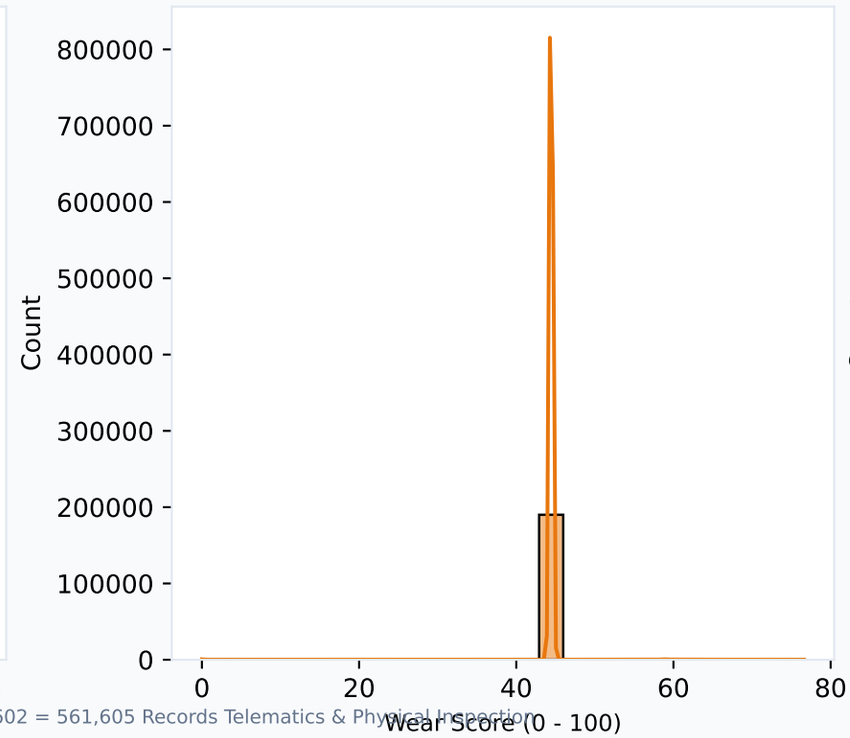
Projected Tyre Life (km) by Axle



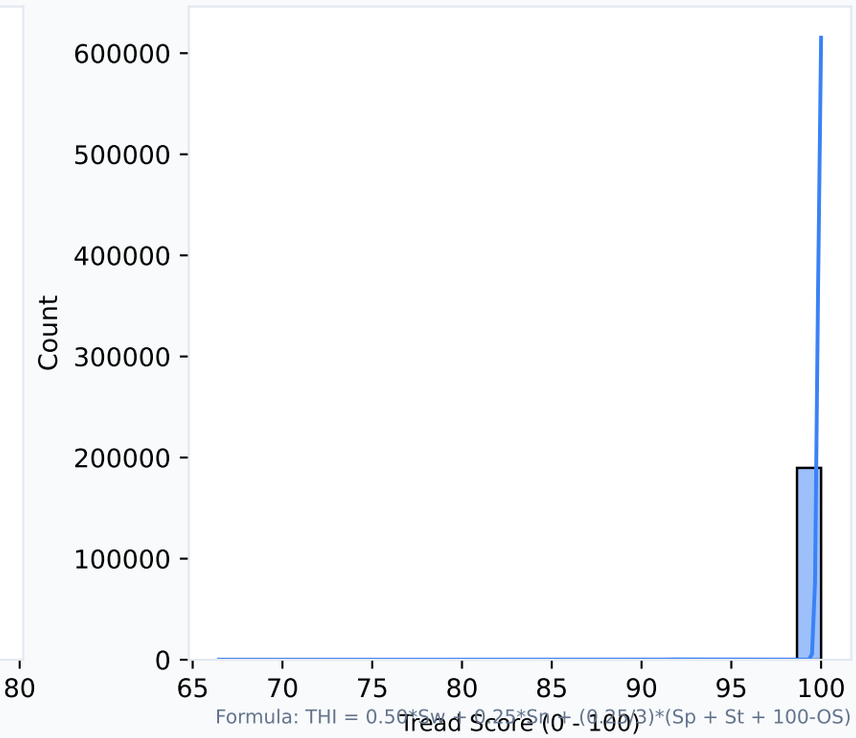
Groove-to-Groove Wear Asymmetry



Wear Score (Sw) Component Distribution



Tread Score (Sn) Component Distribution

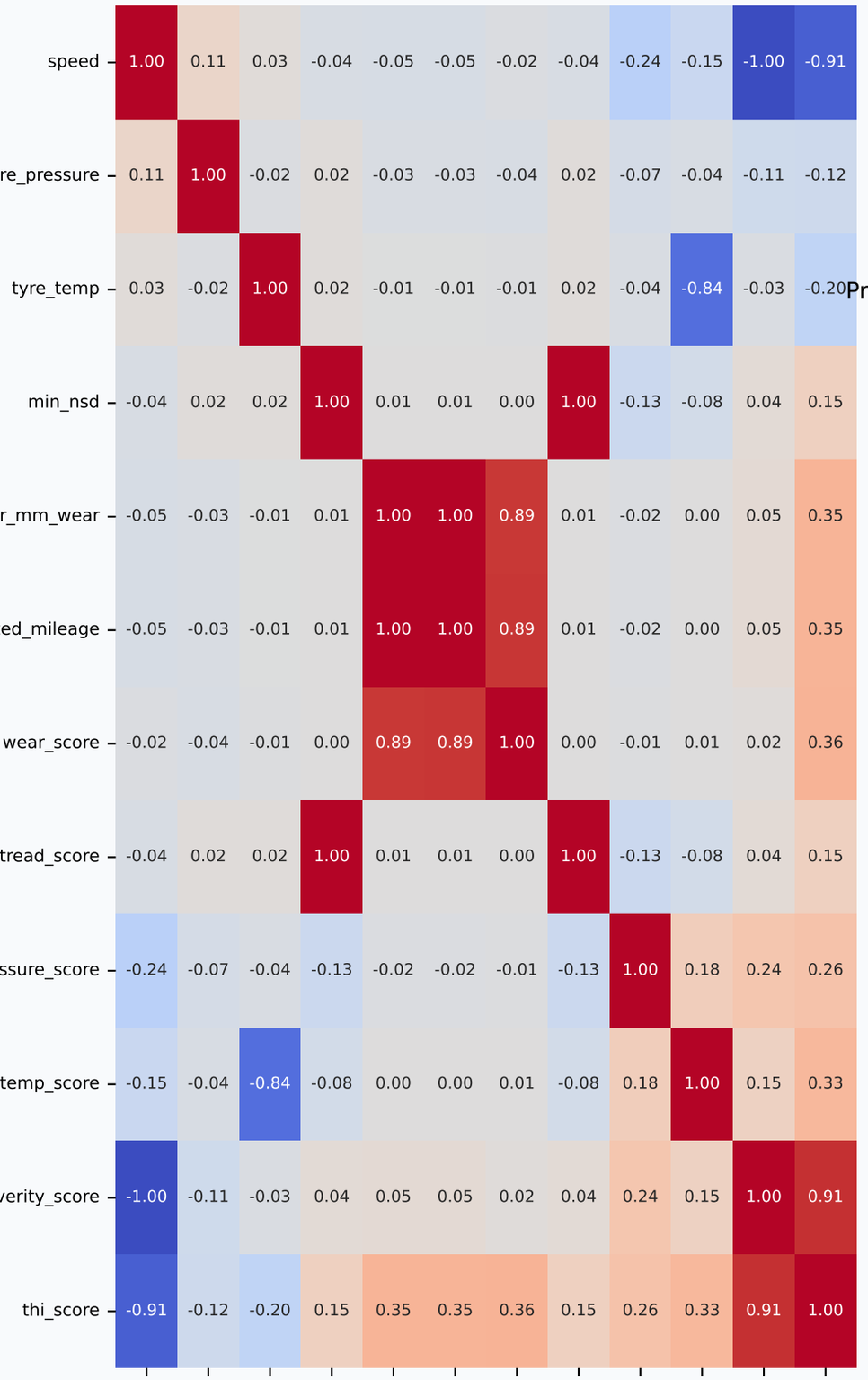


Formula: THI = 0.50*Sw + 0.25*Sn + (0.25/3)*(Sp + St + 100-OS)

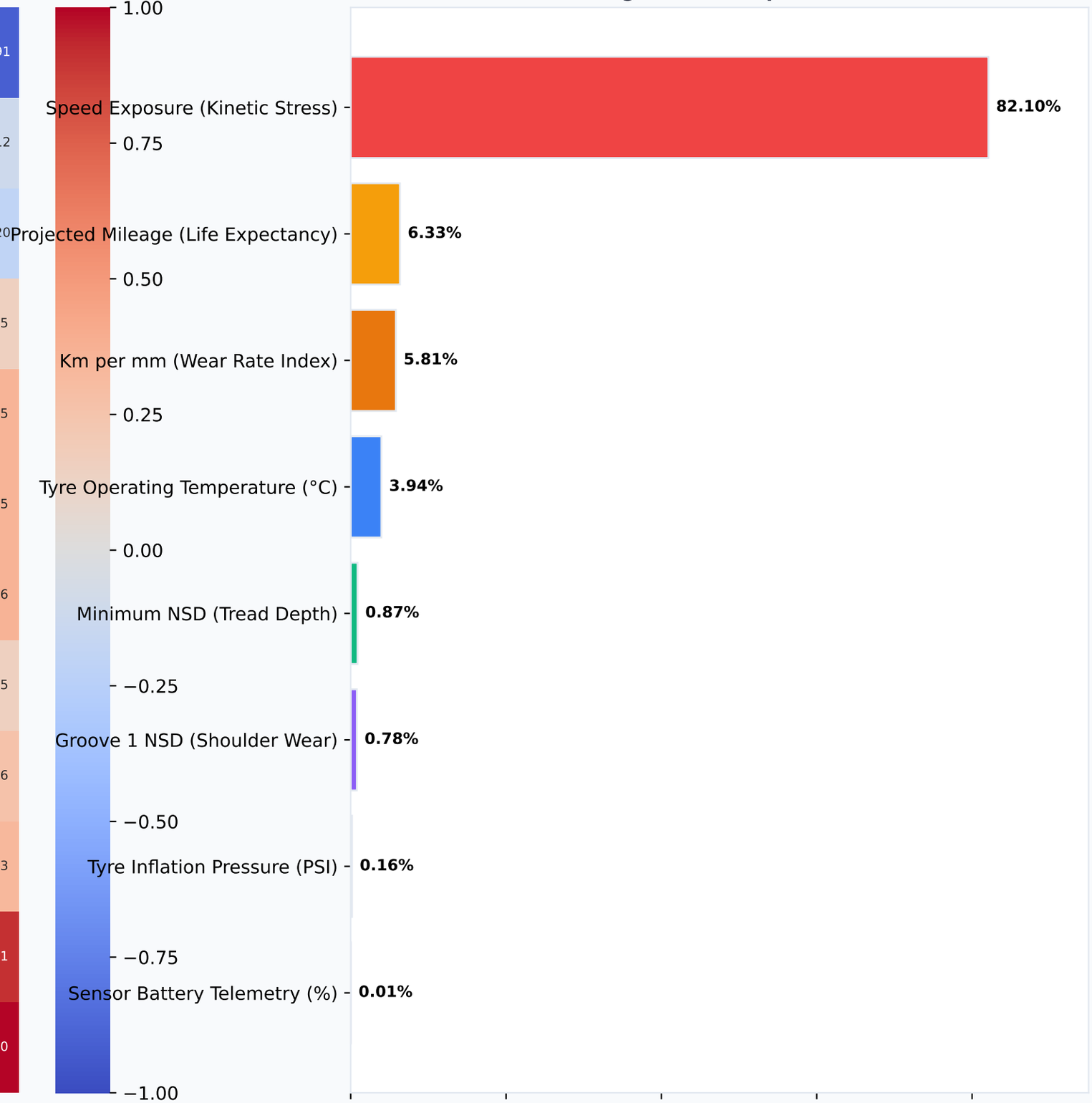
CORRELATION MATRIX & SENSOR INTERPLAY ARCHITECTURE

Pearson Multicollinearity Analysis & Operational Sensitivity with Tyre Health Index

Full Pearson Correlation Heatmap



Gradient Boosting Feature Importance Distribution

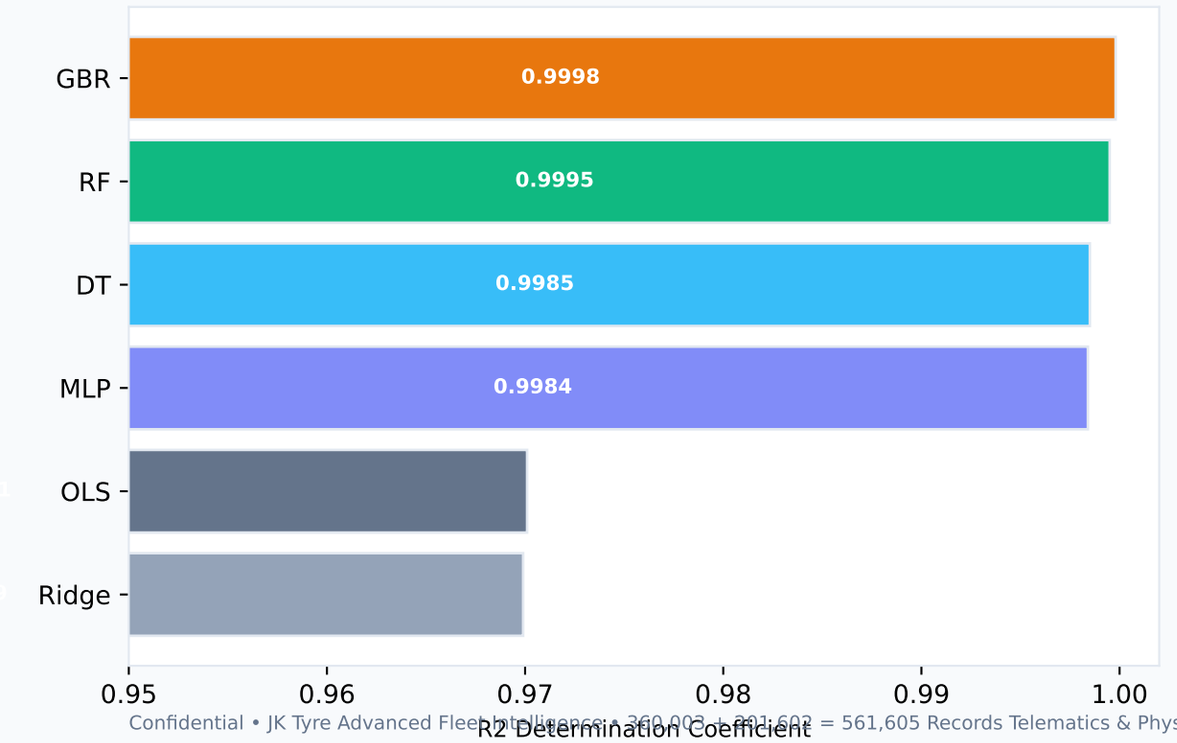


Formula: THI = 0.50*Sw + 0.25*Sn + (0.25/3)*(Sp + St + 100-OS)

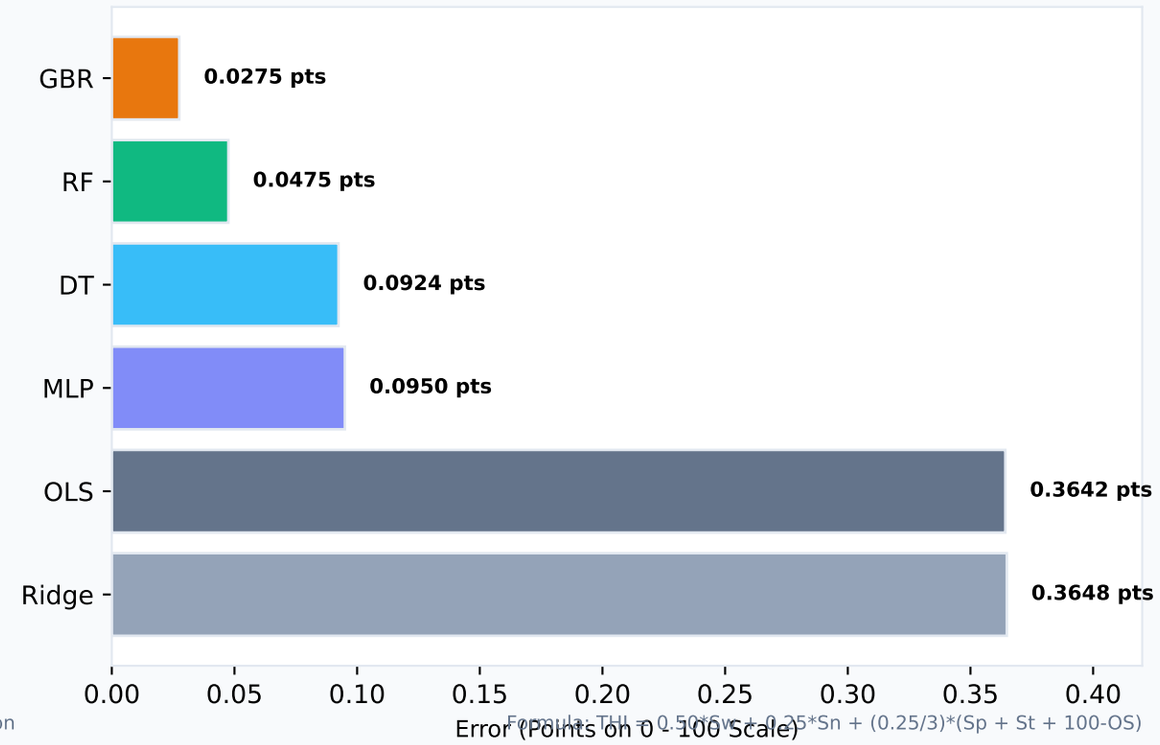
Machine Learning Model Tournament Leaderboard

Rank	Model Architecture	Test R2	MAE (pts)	RMSE (pts)	MAPE	3-Fold CV R2	Trip 4 Holdout	Fit Time	Status
1	Gradient Boosting Regressor	0.9998	0.0275	0.0462	0.0409%	0.9932	0.9998	62.3s	CHAMPION
2	Random Forest Regressor	0.9995	0.0475	0.0745	0.0706%	0.9908	0.9993	13.5s	RUNNER UP
3	Decision Tree Regressor (	0.9985	0.0924	0.1302	0.1371%	0.9896	0.9972	1.5s	FAST TREE
4	Multi-Layer Perceptron (M	0.9984	0.0950	0.1371	0.1396%	0.9433	0.9965	91.1s	NEURAL
5	Linear Regression (OLS	0.9701	0.3642	0.5859	0.5336%	0.9642	0.9680	1.1s	LINEAR
6	Ridge Regression (L2)	0.9699	0.3648	0.5879	0.5349%	0.9642	0.9678	0.1s	REGULARIZED

Test R2 Score Across Algorithms



Mean Absolute Error (MAE in Points, Lower = Better)



DYNAMIC MULTI-AXLE FLEET CONFIGURATION SUPPORT (CONFIGURATION-AGNOSTIC)

Scalable THI Evaluation from 4-Wheel LCVs to 18-Wheel Multi-Axle Tractor-Trailers

Supported Commercial Vehicle Configurations

Layout	Axles	Active Wheel Positions	Commercial Application
4-Wheel (4x2)	2 Axles	1L, 1R, 2L, 2R	Delivery LCV, Passenger Van
6-Wheel (4x2 Dual)	2 Axles	1L, 1R, 2LO, 2LI, 2RO, 2	Medium Commercial Vehicle
10-Wheel (6x4)	3 Axles	Steer (2) + Dual Tanden	Heavy Tipper, Bulk Hauler
12-Wheel (8x2)	4 Axles	Steer (2) + Drive (2) + T	Multi-Axle Cargo Truck (Audit Truck)
14-Wheel (10x2)	5 Axles	Steer (2) + Lift (2) + Tar	Heavy Cement / Steel Transporter
18-Wheel (Puller)	5+ Axles	Tractor (10) + Semi-Trail	Long-Haul Inter-State Container

Configuration-Agnostic Mathematical Architecture

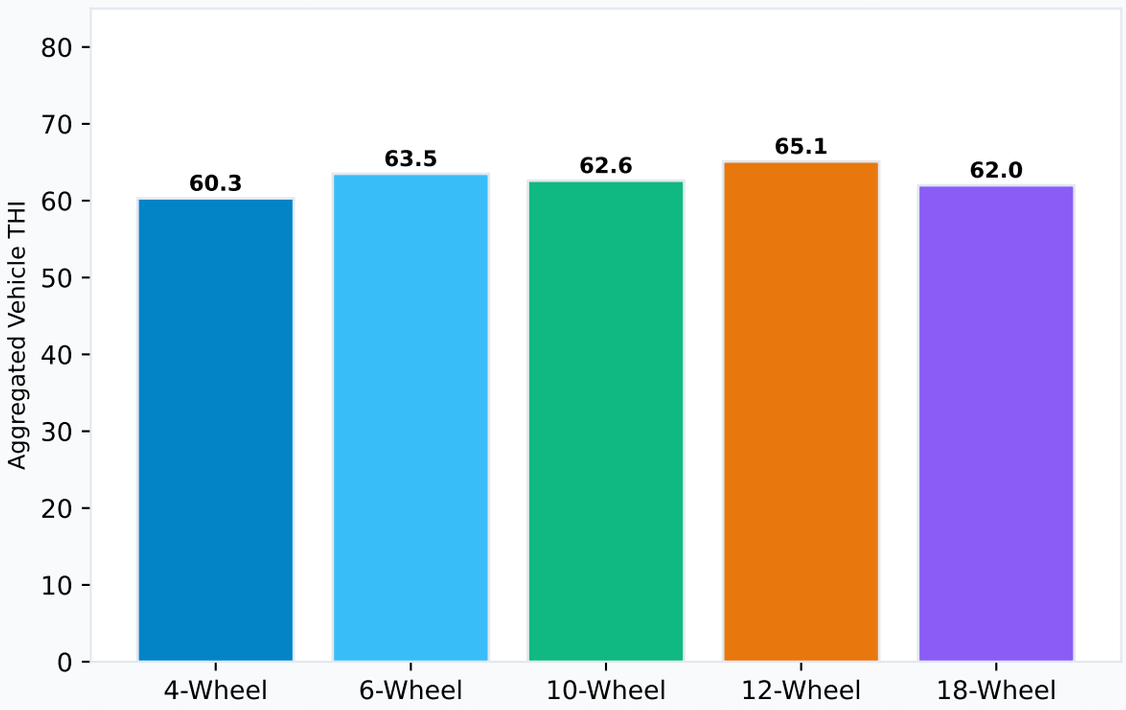
```
1. Intrinsic Per-Tyre Independence:
   Each tyre's THI is evaluated individually based on its own telematics (P, T),
   inspection tread depth (NSD, km/mm wear), and axle role:
   THI_i = 0.50*Sw_i + 0.25*Sn_i + (0.25/3)*(Sp_i + St_i + 100 - OS)

2. Safety-Critical Axle Weighting:
   Steer Axle:     w_steer   = 1.20 (Critical directional safety)
   Drive Axle:     w_drive   = 1.00 (Traction & torque stress)
   Tandem / Tag:   w_tandem  = 0.90 (Load bearing suspension)
   Trailer Axle:   w_trailer = 0.85 (Trailing stability)
   Lift Axle:      w_lift    = 0.80 (Retractable auxiliary)

3. Dynamic Vehicle Aggregation Formula (N wheels):
   THI_vehicle = [ sum(w_i * THI_i) ] / [ sum(w_i) ]

4. Weakest-Link Alert Protection:
   THI_min = min(THI_i for i in 1..N)
   If THI_min < 70, vehicle triggers immediate operational warning!
```

Empirical THI Across Tested Vehicle Layouts



Operational Thresholds & Action Matrix

Score Range	Status	Color Code	Operational Action Guidelines
90 – 100	EXCELLENT	#10B981	Nominal wear, optimal pressure & t
85 – 89	GOOD	#3B82F6	Normal road operations. Standard s
70 – 84	WARNING	#F59E0B	Elevated wear or pressure deviation
< 70	CRITICAL	#EF4444	Severe wear, high thermal risk or le